Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY **9700/42**

Paper 4 A Level Structured Questions **October/November 2021**

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6	Calculation specific guidance Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	Guidance for chemical equations Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations:

;	separates marking points
/	alternative answers for the same marking point
R	reject
A	accept
I	ignore
AVP	any valid point
AW	alternative wording (where responses vary more than usual)
ecf	error carried forward
underline	actual word underlined must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>
<table>
<tr><th>feature</th><th>letter</th></tr>
<tr><td>glomerular filtrate</td><td>E ;</td></tr>
<tr><td>basement membrane</td><td>C ;</td></tr>
<tr><td>podocyte cells</td><td>D ;</td></tr>
<tr><td>capillary endothelial cell</td><td>A ;</td></tr>
</table>
</td><td>4</td></tr>
<tr><td>1(b)</td><td>any five from:

1 afferent arteriole, (lumen) wider / has larger diameter, than efferent ;

2 causes high, blood / hydrostatic, pressure in, glomerulus / capillaries ;

3 pores / gaps / fenestrations, in capillary endothelium ; I holes in cells

4 basement membrane acts as a filter / AW ;

5 detail of podocytes ; e.g. finger-like structures / folds / slit pores

6 correct ref. to passage of substances ; e.g. basement membrane stops, blood cells / large proteins / molecules above RMM 68 000–70 000 or allows urea / glucose / amino acids / ions / water

7 idea of network of capillaries ;</td><td>5</td></tr>
<tr><td>1(c)(i)</td><td>any two from:

1 as (mean arterial blood) pressure increases the GFR increases, levels off and increases / AW ;

2 (plateau occurs) at 100 $cm^3\ min^{-1}$ / between 10 and 20 kPa ;

3 AVP ; e.g. plateau is normal or healthy value</td><td>2</td></tr>
</table>

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Question	Answer	Marks
1(c)(ii)	*any **one** from:* 1 kidney, damage / disease / cancer **;** 2 dehydration **;** 3 low, blood / hydrostatic, pressure **or** blood loss **;** 4 AVP **;** e.g. other (relevant) disease	**1**

Question	Answer	Marks
2(a)	*any **one** from:* 1 industrialisation **;** 2 transportation **;** 3 power stations **;** 4 burning of fuels **;**	**1**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(b)</td><td>

*any **three** from:*

1 genetically modified has a higher (dry) mass throughout the experiment **; ora**

2 genetically modified has a (slightly) higher rate of / steeper, increase (in dry mass) **; ora**

3 the largest difference in (dry) mass is after day 8 **;**

4 paired data quote with units **;**

<table>
<tr><th rowspan="2">time / days</th><th colspan="3">dry mass / g dm^{-3}</th></tr>
<tr><th>genetically modified ± 0.02</th><th>unmodified ± 0.02</th><th>difference between the two ± 0.04</th></tr>
<tr><td>2</td><td>0.22</td><td>0.13</td><td>0.09</td></tr>
<tr><td>4</td><td>0.48</td><td>0.34</td><td>0.14</td></tr>
<tr><td>6</td><td>0.62</td><td>0.48</td><td>0.14</td></tr>
<tr><td>8</td><td>0.88</td><td>0.74</td><td>0.14</td></tr>
<tr><td>10</td><td>1.24</td><td>0.96</td><td>0.28</td></tr>
<tr><td>12</td><td>1.54</td><td>1.31</td><td>0.23</td></tr>
<tr><td>14</td><td>1.86</td><td>1.52</td><td>0.34</td></tr>
</table>

rate of growth over 14 days
genetically modified = 1.86 / 14 = 0.13 g dm^{-3} day^{-1}
unmodified = 1.52 / 14 = 0.11 g dm^{-3} day^{-1}

</td><td>**3**</td></tr>
</table>

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Question	Answer	Marks
2(c)	*any **three** from:* 1 (more rubisco so) greater rate of / **more**, carbon (dioxide) fixation **;** AW 2 (so) greater rate of **/ more**, GP produced **;** 3 so greater rate of / **more** TP, produced (from GP) **;** 4 (so) greater rate of / **more**, regeneration of RuBP **;** 5 (so) greater rate of **/ more**, Calvin cycle **;** 6 AVP **;** e.g. carbon fixation is a rate-limiting step / concentration of rubisco is a limiting factor	**3**
2(d)	*any **three** from:* 1 glucose for respiration **;** 2 starch for storage **;** 3 cellulose to make cells walls **;** 4 sucrose for, translocation / described **;** 5 fatty acids and glycerol / lipid, to make membranes **or** fatty acids and glycerol / lipid / fats, for storage **or** fatty acids to make acetyl CoA (for Krebs cycle) **;** 6 amino acids to make, proteins / enzymes **;** 7 proteins for, growth / repair **;**	**3**

Question	Answer	Marks
2(e)	*any **one** from:* 1 *idea of* fast(er) growth rate / reproduction ; 2 quicker to set up / AW ; 3 cheaper to set up / AW ; 4 (as aquatic) not using (useful) land ; 5 can, culture / grow, algae in the lab ;	**1**

Question	Answer	Marks
3(a)	*any **three** from:* 1 seed / embryo, absorbs water ; 2 (stimulates) embryo to produce gibberellin ; 3 gibberellin moves to aleurone layer ; 4 gene coding for amylase, expressed / transcribed ; 5 translation of mRNA (to produce amylase) ; 6 AVP ; e.g. gibberellin binds to receptor / ref. destruction of DELLA protein	**3**
3(b)(i)	endosperm ;	**1**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>3(b)(ii)</td><td>
any four from:

1 amylase activity increases ;

2 as (seeds) produce more amylase ;

3 percentage of starch decreases ;

4 amylase, hydrolyses / breaks down, starch ;

5 idea of a change at an increasing rate (for either) ; A exponential change

6 data quote: amylase or starch at two different days ;

<table>
<tr><th>time / days</th><th>amylase activity / au</th><th>percentage of starch remaining</th></tr>
<tr><td>0</td><td>0.0</td><td>100</td></tr>
<tr><td>1</td><td>0.2</td><td>99</td></tr>
<tr><td>2</td><td>0.8</td><td>98</td></tr>
<tr><td>3</td><td>2.4</td><td>95</td></tr>
<tr><td>4</td><td>5.0</td><td>91</td></tr>
<tr><td>5</td><td>8.4</td><td>82</td></tr>
</table>
accept manipulated data
</td><td>4</td></tr>
</table>

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Question	Answer	Marks
3(c)(i)	*any **three** from:* 1 enzyme, denatured / deactivated / inactivated **;** 2 changes, tertiary structure / 3D shape, of amylase **; A** enzyme 3 changes shape of active site **;** 4 active site, no longer complementary / does not bind, to starch / substrate **or** no, ESC / enzyme-substrate complexes, formed **;** 5 AVP **;** e.g. removes water from embryo for gibberellin synthesis / removes water from endosperm for hydrolytic reactions	**3**
3(c)(ii)	*any **one** from:* 1 stops, sugars / glucose, from being used **; ora** 2 ensures (enough) sugars for beer production **;**	**1**

Question	Answer	Marks
4(a)(i)	*any **four** from:* 1 obtain single-stranded DNA **;** 2 label DNA with fluorescent, dye / tag / label **;** 3 probes on, chip / microarray **;** 4 each probe is unique to a particular, SNP / gene / allele **;** 5 DNA (added), binds / hybridises, to probes **;** 6 wash off excess DNA (after hybridisation) **;** 7 (view under) UV light / laser scanner / high resolution digital camera **;** 8 fluorescence indicates presence of, SNP / gene **; R** gene expressed	**4**
4(a)(ii)	*any **two** from:* 1 large database **;** 2 use database to find, SNPs / probes **;** 3 fast / accurate / efficient **;** 4 *ref. to* (computer) software / algorithms **;** 5 *ref. to* statistics / statistical analysis **;** 6 *idea of*, $17\,000 \times 500\,000$ / 8.5×10^9, data points **;**	**2**
4(b)(i)	1 and 6 **;**	**1**

Question	Answer	Marks
4(b)(ii)	*any **two** from:* 1 all diseases, are autosomal / not associated with X chromosome / not sex-linked **;** 2 all diseases are associated with SNP on chromosome 6 **;** 3 type 1 and type 2 diabetes SNPs are associated with chromosomes 12 and 16 (but not rheumatoid arthritis) **;** 4 type 2 diabetes SNP associated with chromosome 10 (but not type 1 diabetes and rheumatoid arthritis) **;** 5 arthritis and type 1 diabetes have higher level of association (with each other) than with type 2 diabetes **;** 6 rheumatoid arthritis has fewest number of SNPs **or** type 1 and type 2 diabetes has more SNPs (than rheumatoid arthritis) **;**	**2**

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Question	Answer	Marks
4(c)	*any **three** from:* 1 DNA analysis not available for everyone **;** 2 lifestyle change **; A** e.g. lose weight / stop smoking / take exercise 3 early treatment **;** 4 allows people to plan **;** e.g. organise care / appoint power of attorney / take out medical insurance / make will / gift property to children / retire early 5 decide whether to have children / ref. (therapeutic) termination **;** 6 results may cause, anxiety / stress / panic / depression (if positive) **or** reduce worry (if negative) **;** 7 results may affect ability to get, insurance / credit / jobs **;** 8 *idea of* predictions may not be accurate **;**	**3**

Question	Answer	Marks
5(a)	*Fig. 5.1:* <u>species</u> **;** *Fig. 5.2:* <u>genetic</u> **;** *Fig. 5.3:* habitat / ecosystem **;**	**3**

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Question	Answer	Marks
5(b)(i)	*plants:* 1 quadrats **;** 2 transects **;** 3 *ref. to* use, ACFOR / Braun-Blanquet **or** estimate percentage cover (of species) count species calculate plus species abundance record species density AW species frequency **;** *insects:* 4 pooters / pitfall traps / nets / light traps / sugar traps **;** 5 mark-release-recapture **;** *accept in either section:* 6 (identification) keys **;**	**4**
5(b)(ii)	*estimate biodiversity:* Simpson's index (of biodiversity) **;** *test relationship:* Pearson's linear correlation coefficient / Spearman's rank correlation **;**	**2**

Question	Answer	Marks
5(c)	*any **three** from:* *negative* 1 smaller crop area so lower yield **;** 2 prairie strip insects may feed on crop **;** 3 (so) less, profit / income **;** *must be linked to mp1 or mp2* *positive* 4 (soil) more fertile / described, so higher yield **;** 5 less / no, fertiliser / phosphorus compounds / nitrogen compounds, used **; ignore** minerals 6 (so) more, profit / income **or** reduction in expenses **;** *must be linked to mp4 or mp5*	**3**

Question	Answer	Marks
6(a)(i)	*any **four** from:* 1 water soluble **;** 2 (so) easily transported around the cell **;** 3 (ATP) loses, P_i / phosphate group **;** 4 hydrolysed by ATPase / (ATP is) stable molecule **;** 5 to release energy, immediately / in small packets **or** *ref. to* 30.5 kJ (mol^{-1}) **;** 6 can be, recycled / regenerated **or** ATP ⇄ ADP + P_i **;**	**4**
6(a)(ii)	*any **two** from:* 1 unwinding DNA **;** 2 activating (RNA) nucleotides **;** 3 *ref. to* mRNA synthesis **;** 4 peptide bond formation / joining amino acids **;** 5 AVP **;** e.g. post translation modification / amino acid activation / moving ribosomes along mRNA / movement of mRNA from nucleus	**2**

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Question	Answer	Marks
6(b)(i)	<table><tr><th>statement</th><th>letter</th></tr><tr><td>the site of the Krebs Cycle</td><td>**A**</td></tr><tr><td>a phospholipid bilayer impermeable to H+ ions</td><td>**E / D**</td></tr><tr><td>the site of translation</td><td>**C**</td></tr></table> ;; *all correct = 2 marks* *1 or 2 correct = 1 mark*	**2**
6b(ii)	$(2 \times \pi \times 0.36^2) + (2 \times \pi \times 0.36 \times 5.25)$ **or** $2\pi \times 0.36\ (0.36 + 5.25)$ **;** 12.7 **or** 12.68 **or** 12.69 **;** *accept 3.14 or π in workings*	**2**
6(b)(iii)	1 (more cristae results in) **more**, ETC / (named) carrier proteins / ATP synthase **;** **or** (more cristae results in) **more**, oxidative phosphorylation / chemiosmosis / ATP synthesis **;** 2 (because) cardiac muscle must undergo continuous contractions / AW **;**	**2**

Question	Answer	Marks
7(a)	*any **two** from:* 1 tyrosine converted to, DOPA / dopaquinone **;** 2 dopaquinone converted to melanin **;** 3 *ref. to* melanocytes **;**	**2**
7(b)(i)	parental phenotype (black x yellow) parental genotype **BbEe x Bbee ;** gametes **BE Be bE be Be be ;** offspring genotypes **;; BBEe BbEe BbEe BBee Bbee Bbee bbee bbEe** offspring phenotypes **; black black black yellow yellow yellow yellow brown** ratio 3 black : 1 brown : 4 yellow **;** *ecf to max 3*	**6**
7(b)(ii)	transcription factor **;**	**1**

Question	Answer	Marks
8	*any **five** from:* 1 geographical, isolation / separation **or** *ref. to* Panama / land, separating populations **;** 2 no, gene flow / interbreeding, between both populations **;** 3 different, environmental conditions / selection pressures **;** 4 random / different, mutations **;** 5 (different), alleles selected for / gene pool / changes in allele frequency **;** 6 over time populations have different, morphological / physiological / behavioural, features **;** 7 eventually reproductive isolation occurs **;** 8 allopatric speciation **;**	**5**

Question	Answer	Marks
9(a)	*any **nine** from:* 1 hypothalamus / osmoreceptors, detects changes in water potential of blood ; 2 (causes) posterior pituitary gland ; 3 (to) release ADH into blood ; 4 ADH binds to receptors ; 5 on cell surface membrane of collecting duct (cells) ; 6 stimulates enzyme cascade / phosphorylase enzyme produced ; 7 vesicles, move towards / fuse with, cell surface membrane ; 8 (vesicles have) aquaporins ; 9 collecting duct, membranes / cells / walls, more permeable to water ; 10 water moves, down water potential gradient / by osmosis ; 11 into, (collecting duct) cells / tissue fluid / blood ; 12 water potential (of blood) returns to set point ;	**9**

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Question	Answer	Marks
9(b)	*any **six** from:* 1 dendrites attached to cell body **;** 2 nucleus in, cell body / soma **;** 3 many mitochondria **;** 4 much rough endoplasmic reticulum / Nissl's granules (in cell body) **;** 5 long axon **;** 6 synaptic knobs / synaptic bulbs / terminal branches / axon terminals **;** 7 Schwann cells / myelin sheath / myelinated **;** 8 nodes of Ranvier **;** 9 cell body in, CNS / brain and spinal cord **;** *accept from labelled diagram* *if sensory neurone described – max 5*	**6**

Question	Answer	Marks
10(a)	*any **seven** from:* 1 (dip stick) has immobilised enzymes **;** 2 (dip stick) dipped in urine **; A** person, urinates / AW, on stick 3 glucose oxidase **and** peroxidase **;** 4 glucose reacts to give hydrogen peroxide **;** 5 (hydrogen peroxide reacts with) colourless substance / chromogen **;** 6 to give, colour change / coloured substance **;** 7 compare with colour chart **;** 8 more glucose gives darker colour / colour intensity gives glucose quantity **;** 9 AVP **;** e.g. doesn't give current blood glucose concentration / not numerical / semi-quantitative	**7**

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Question	Answer	Marks
10(b)	*any **eight** from:* 1 dominant allele / *Le*, codes for, functional enzyme **; ora** 2 (enzyme) produces active gibberellin (GA) **;** 3 DELLA (protein) inhibits, transcription factor / PIF **or** DELLA (protein) prevents transcription **;** 4 gibberellin / GA, binds to receptor (complex) **; ignore** cell surface membrane 5 *ref. to* enzyme involved **;** 6 causes DELLA (protein) destruction **; R** GA breaks DELLA (protein) 7 transcription factor / PIF / RNA polymerase, binds to, DNA / promoter **;** 8 (growth) genes, switched on / expressed / transcribed **or** transcription occurs **;** 9 cell walls loosen / acid growth (described) **;** 10 (so) cells can expand when water enters **;** 11 *ref. to* **cell**, elongation / division **;** 12 increases internode length **;** 13 AVP **;** e.g. *ref. to* expansins / interaction with auxin / *ref. to* XET	**8**

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